Beyond Pairwise GUI Quality Judgments: Multi-Dimensional Evaluation with Modern Multimodal LLMs

Xinyang Xie

Department of Informatics, Technische Universitaet Clausthal, Clausthal-Zellerfeld, Germany.

Abstract

Assessing the quality of graphical user interfaces (GUIs) is becoming increasingly important as large language models are used to generate interface designs and frontend code. Existing screenshot-based approaches can support GUI quality judgments, but they remain limited by binary decisions, older models, and simple prompting strategies. This proposal extends that line of work with current multimodal large language models. The study combines pairwise comparison with rubric-based assessment across layout quality, visual hierarchy, information organization, and perceived usability. Methodologically, it uses a screenshot-based evaluation set, a human annotation protocol with agreement analysis, and a comparison of judging strategies such as zero-shot prompting, rubric-guided scoring, repeated judging, and order-swapped pairwise comparisons, where the order of candidates is reversed to detect positional bias. The resulting protocol will be packaged into a benchmark and lightweight leaderboard that can be extended to newly added models. In addition to the static screenshot-based track, the proposal includes a small dynamic evaluation module in which computer-use agents attempt short tasks on selected GUIs. The goal is to determine which evaluation setup aligns best with human judgments and how static quality judgments relate to dynamic interaction performance.

1 Introduction and Motivation

The quality of a graphical user interface (GUI) shapes how efficiently users complete tasks and how they perceive a software system. As generative models are increasingly used to produce interface mockups and frontend code, the question of how UI quality can be evaluated becomes more pressing. Importantly, GUI quality is not limited to visual appearance alone: it also concerns whether an interface supports successful interaction and task completion once users begin to navigate it.

UIClip [1] provided an important first step by showing that vision-language models can perform pairwise comparisons of UI screenshots and identify which of two interfaces appears better. This screenshot-based setting is well suited for evaluating visual structure and perceived usability, but it does not capture whether an interface also supports reliable task completion in interactive use. For that aspect, dynamic assessment methods, such as computer-use agents that can interact with interfaces automatically, are particularly relevant.

Recent work on improving UI generation from designer feedback [2] demonstrates that high-quality, workflow-aligned feedback can be far more informative than large amounts of coarse ranking data; notably, simple pairwise rankings in that study reached only around 50% evaluator agreement, close to chance level. These findings suggest that GUI quality assessment benefits from richer evaluation structures – such as rubric-based scoring and requirement-aligned judgment – rather than binary preference alone. In parallel, the broader LLM-as-a-judge literature [3] has introduced prompting, consistency, and aggregation strategies that may also be useful for GUI evaluation. Taken together, these developments motivate a more methodologically explicit re-evaluation of GUI quality judgment with contemporary multimodal models.

This thesis therefore asks whether modern multimodal large language models can approximate human judgments of GUI quality beyond pairwise comparison, and how such judgments should be elicited and evaluated in a reliable way. Its main focus is a benchmark and leaderboard for GUI quality evaluation built around two complementary tracks: a screenshot-based baseline track and a requirement-driven generation track. This benchmark is complemented by a small dynamic evaluation module in which computer-use agents examine task completion on selected interfaces.

2 Planned Approach

The planned work centers on an evaluation pipeline with six main components: dataset construction, human annotation, rubric design, the comparison of LLM-as-a-judge strategies, benchmark and leaderboard design, and a small dynamic evaluation extension.

2.1 Datasets and Evaluation Material

The evaluation material will be organized into two complementary tracks. The first track draws on publicly available screenshot-based resources from UIClip as an established baseline. These existing screenshots will be used for pairwise comparison and rubric-based assessment of visual design quality. To avoid evaluating only older interfaces, this material may be supplemented by a smaller curated set of contemporary web or mobile screenshots.

The second track will be constructed around natural language requirements. A set of short UI descriptions will be collected or designed, specifying the intended functionality, layout, and content of an interface. For this track, a small set of interfaces will be generated from shared natural language descriptions. Recent work has shown that current multimodal LLMs can generate renderable front-end code from visual and textual UI specifications [4], making this pipeline practically feasible. The resulting front-end code will then be rendered as screenshots for visual evaluation, while remaining executable for the dynamic evaluation module. Because this track produces runnable front-end code rather than static images alone, it supports both browser-based interaction tests and checks of whether the generated interface satisfies the original requirements.

Together, the two tracks allow the benchmark to cover both existing, human-designed interfaces and newly generated ones, broadening the range of evaluation scenarios beyond what a screenshot-only dataset would permit.

2.2 Human Annotation Protocol

Human judgments will be collected for a selected evaluation subset. For pairwise tasks, annotators will compare two GUI screenshots and indicate which interface is better overall, with an optional tie label if required. For single-image tasks, annotators will score each GUI with a small rubric and provide brief justifications. To improve the reliability of the reference labels, each item will be annotated by multiple raters, and inter-rater reliability will be measured, for example with Krippendorff's alpha.

2.3 Multi-dimensional GUI Rubric

To move beyond binary comparison, I will define a small rubric grounded in common visual design principles and usability heuristics. The initial dimensions are:

- **Layout quality**: spatial organization, alignment, spacing, and overall balance.
- **Visual hierarchy**: whether contrast, size, and color guide attention appropriately.
- **Information organization**: clarity of grouping and logical ordering of content.
- **Perceived usability**: how easily key actions and interface purpose can be inferred from a static screenshot.
- **Requirement fidelity** (Track B only): the degree to which the generated interface realizes the elements, structure, and functionality specified in the original natural language description.

Each dimension will be scored on a fixed ordinal scale with short written anchors so that the same rubric can be applied by both human annotators and model judges. The perceived usability dimension is intentionally limited to what can reasonably be inferred from a static screen, rather than to full interaction quality. The requirement fidelity dimension applies exclusively to the requirement-driven track, where an original description is available as a reference; for this dimension, judges compare the rendered interface against the description and assess how completely and accurately the requirements have been met.

2.4 Prompting and Judging Strategies

Several judging strategies will be compared. A zero-shot baseline will ask the model directly for a pairwise preference or a rubric-based score. A few-shot variant will add one or two worked examples to test whether light calibration improves judgment quality. A rubric-guided condition will provide explicit scoring instructions and require a structured output format. To test robustness, the same prompt will be repeated to measure self-consistency. For pairwise tasks, order-swapped pairwise comparisons will be used, where the order of candidates is reversed to detect positional bias. In addition, the study will compare single-model judgments with simple aggregation strategies such as majority voting for pairwise tasks and averaged scores for rubric-based tasks.

2.5 Benchmark and Leaderboard Design

Beyond comparing individual models, the project will package the evaluation setup into a benchmark structure that combines pairwise tasks, rubric-based scoring, and shared reference annotations. Results will be stored in a standardized format so that newly added multimodal models can be evaluated under the same protocol. As a practical artifact, the benchmark will be accompanied by a lightweight leaderboard or web-based dashboard for comparing models across tasks and rubric dimensions.

2.6 Dynamic Evaluation with Computer-Use Agents

To complement the static benchmark, the project will include a small dynamic evaluation extension. This module will operate primarily on the requirement-driven track, where LLM-generated HTML interfaces can be rendered and executed in a browser. For a selected subset of these interfaces, short interaction tasks derived from the original natural language descriptions will be defined and executed by computer-use agents. Prior work has demonstrated that multimodal agents can perform goal-directed interaction tasks on visual web interfaces [5], providing a methodological foundation for this module. This extension will be kept deliberately small and will focus on whether interfaces that receive strong static quality judgments also support reliable task completion in practice, and whether requirement alignment in the static setting predicts successful interaction in the dynamic setting.

3 Evaluation

Several current multimodal models will be tested as judges, including representative models from major closed-source and open multimodal model families, depending on availability at evaluation time. The pairwise setting will be evaluated through agreement with human majority labels, while the rubric-based setting will be evaluated through rank correlation and score differences between model outputs and human ratings. For the requirement-driven track, an additional evaluation dimension will measure how well model judgments capture whether a generated interface satisfies its original natural language description. The dynamic extension will be evaluated through task completion rate, action efficiency, and qualitative failure analysis on the selected interactive tasks.

Beyond task accuracy, the evaluation will explicitly examine reliability. This includes inter-rater agreement in the human annotations, consistency across repeated model judgments, stability under order-swapped pairwise comparisons, and agreement across different models. Computational cost will also be tracked in order to assess practical trade-offs between evaluation quality and API usage. A further goal is to determine whether the resulting benchmark and leaderboard provide a stable and extensible basis for comparing both current and newly introduced multimodal models.

Finally, a qualitative error analysis will examine typical failure cases, such as cluttered layouts, minimal interfaces, text-heavy screens, and visually appealing but less usable designs. Together, these analyses should clarify how well modern multimodal large language models support UIClip-style evaluation, which rubric dimensions are most reliable, and which judging strategies align best with human assessment.

References

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